

MTH00050 – Combinatorial Mathematics

Lecture 5: Generating functions

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(Tentative) Schedule 2026

No.	Date	Location	Topic
1	13/01 16/01	C24 C24/E302	Introduction, Sets, Proofs
2	20/01 23/01	C24 C24/E302	General counting methods
3	27/01 30/01	C24 C24/E302	Inclusion-exclusion principle
4	03/02 06/02	C24 C24/E302	Recurrence relations
5	03/03 06/03	C24 C24/E302	Generating functions
			Mid-term test

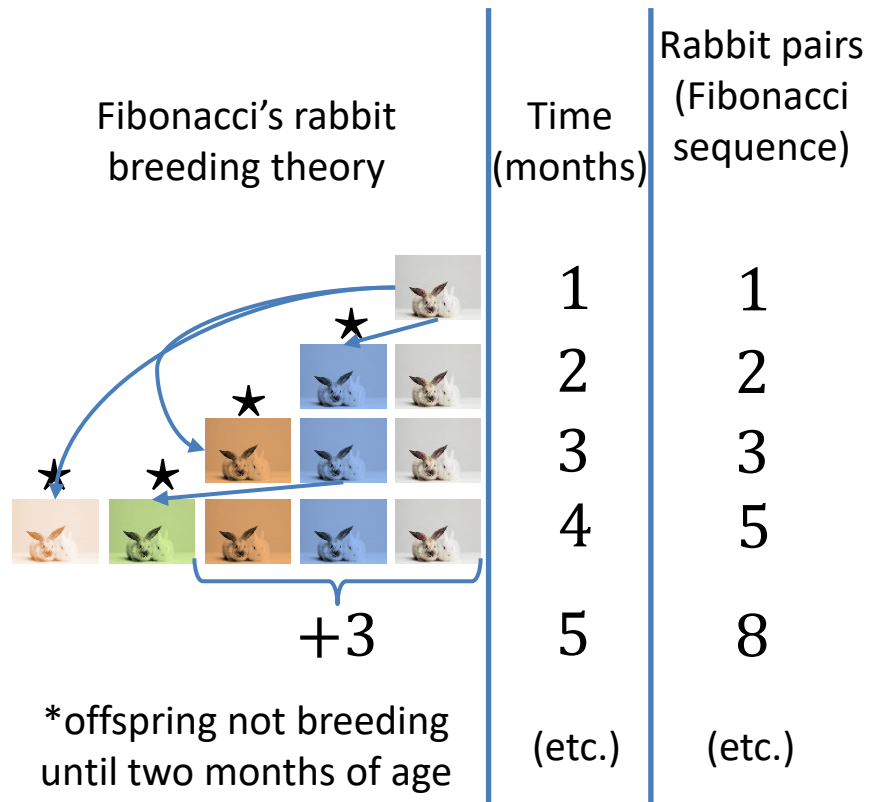
No.	Date	Location	Topic
6	10/03 13/03	C24 C24/E302	Graphs (Part I)
7	17/03 20/03	C24 C24/E302	Graphs (Part II)
8	24/03 27/03	C24 C24/E302	Path problems
9	31/03 03/04	C24 C24/E302	Tree problems
10	07/04 10/04	C24 C24/E302	Network flows (Part I)
11	14/04 17/04	C24 C24/E302	Network flows (Part II)
			Final exam

Goals

1. To help students understand and be able to use generating functions to analyze the complexity of an algorithm.

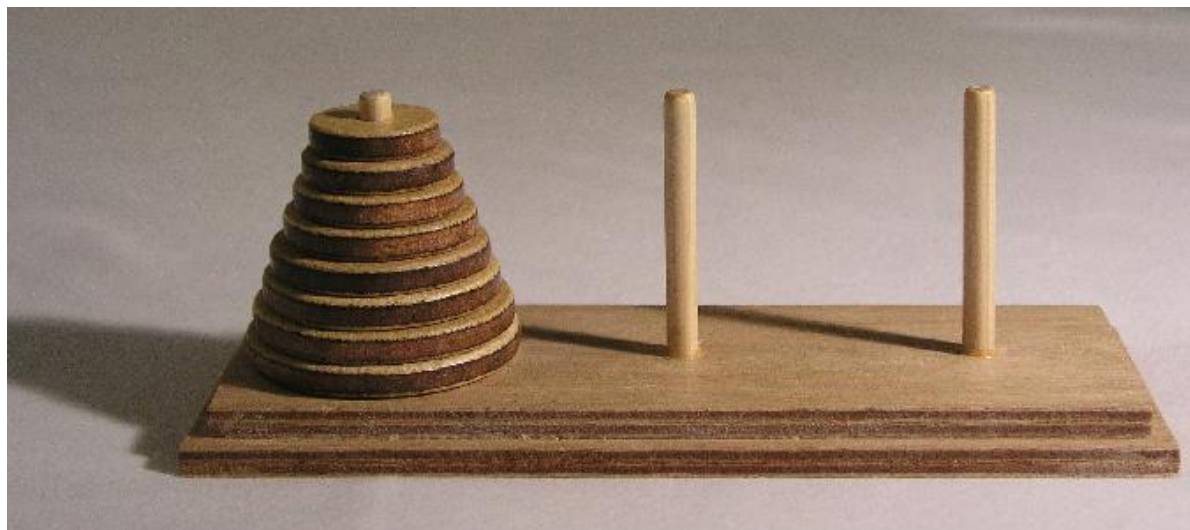
Fibonacci in Animal Breeding Patterns and Hanoi's Tower

How to solve these recurrences without using techniques like the master theorem?



#rabbits at the n th month:

$$T(n) = T(n - 1) + T(n - 2)$$



Number of moves:

$$T(n) = 2T(n - 1) + 1$$

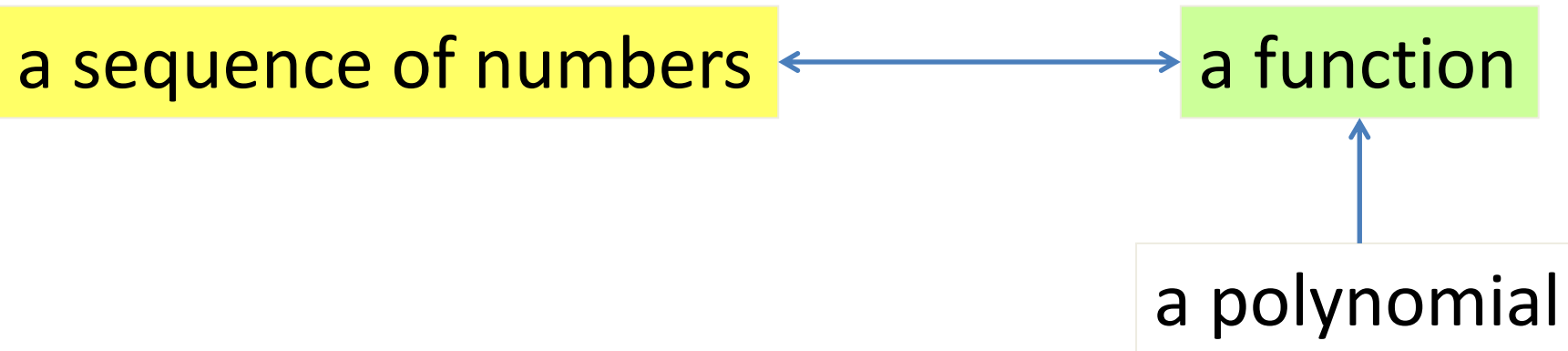
Outline

1. Generating functions for basic sequences
2. Operations on generating functions
3. Counting
4. Solving recurrences

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Abstract



- Through this mapping, we can apply our techniques for manipulating functions.

Ordinary Generating Functions

- Let g_n for $n = 0, 1, \dots$, be a sequence, denoted as $\langle g_0, g_1, g_2, \dots \dots \rangle$.

- The ordinary generating function is:

$$G(x) = \sum_{i=0}^{\infty} g_i x^i = g_0 + g_1 x + g_2 x^2 + \dots$$

- We use a double sided arrow to indicate the correspondence.

$$\langle g_0, g_1, g_2, \dots \dots \rangle \Leftrightarrow g_0 + g_1 x + g_2 x^2 + \dots$$

Simple Examples

- $\langle 0, 0, 0, 0, \dots \rangle \iff 0 + 0x + 0x^2 + 0x^3 + \dots = 0$
- $\langle 1, 0, 0, 0, \dots \rangle \iff 1 + 0x + 0x^2 + 0x^3 + \dots = 1$

The pattern here is simple:

the i th term in the sequence (indexing from 0) is the coefficient of x^i in the generating function.

Geometric Series

- $G_n(x) = 1 + x + x^2 + \dots + x^n + \dots$ ♻
- What is the closed form expression of $G_n(x)$?

$$\begin{array}{rcl} G_n(x) & = & 1 + x + x^2 + \dots + x^n + \dots \\ \underline{xG_n(x)} & = & x + x^2 + \dots + x^n + x^{n+1} + \dots \end{array}$$

$$G_n(x) - xG_n(x) = 1$$

$$G_n(x) = \frac{1}{1-x}$$

More Examples

- $\langle 1, 1, 1, 1, \dots \rangle \Leftrightarrow 1 + 1x + 1x^2 + 1x^3 + \dots = \frac{1}{1-x}$
- $\langle 1, a, a^2, a^3, \dots \rangle \Leftrightarrow 1 + ax + a^2x^2 + a^3x^3 + \dots = \frac{1}{1-ax}$
- $\langle 1, -1, 1, -1, \dots \rangle \Leftrightarrow 1 - 1x + 1x^2 - 1x^3 + \dots = \frac{1}{1+x}$
- $\langle 1, 0, 1, 0, \dots \rangle \Leftrightarrow 1 + 0x + 1x^2 + 0x^3 + \dots = \frac{1}{1-x^2}$

These are all **closed form** generating functions.

Outline

1. Generating functions for basic sequences
2. Operations on generating functions
3. Counting
4. Solving recurrences

manipulations on sequences $\longleftrightarrow$ manipulations on functions

- There are a few basic operations we'll learn.
 - Scaling
 - Addition
 - Right shift
 - Differentiation
 - Product
- We can use these operations to get new sequences from known sequences, and new generating functions from known generating functions.

Scaling

Multiplying a generating function by a constant

→ scales every term in the associated sequence by the same constant.

$$\langle 1, 0, 1, 0, \dots \rangle \Leftrightarrow 1 + 0x + 1x^2 + 0x^3 + \dots = \frac{1}{1 - x^2}$$

Multiply the generating function by 2 gives

$$\frac{2}{1 - x^2} = 2 + 0x + 2x^2 + 0x^3 + 2x^4 + \dots$$

which yields the sequence: $\langle 2, 0, 2, 0, \dots \rangle$

Addition

Adding generating functions corresponds to adding sequences term by term.

$$\begin{array}{rcl}
 \langle 1, 1, 1, 1, 1, 1, \dots \rangle & \Leftrightarrow & \frac{1}{1-x} \\
 + & & \\
 \langle 1, -1, 1, -1, 1, -1, \dots \rangle & \Leftrightarrow & \frac{1}{1+x} \\
 \hline
 \langle 2, 0, 2, 0, 2, 0, \dots \rangle & \Leftrightarrow & \frac{1}{1-x} + \frac{1}{1+x}
 \end{array}$$

The same result as in the previous slide:

$$= \frac{2}{1-x^2}$$

Right shift

$$\langle 1, 1, 1, 1, \dots \rangle \Leftrightarrow 1 + 1x + 1x^2 + 1x^3 + \dots = \frac{1}{1-x}$$

How to generate the sequence $\langle 0, 0, \dots, 0, 1, 1, 1, 1, \dots \rangle$?

k zeros

$$\langle 0, 0, \dots, 0, 1, 1, 1, 1, \dots \rangle \Leftrightarrow x^k + x^{k+1} + x^{k+2} + \dots$$

k zeros

$$= x^k (1 + x + x^2 + \dots)$$

$$= \frac{x^k}{1-x}$$

Adding k zeros $\Leftrightarrow$ multiplying x^k on the generating function.

Differentiation (1/3)

How to generate the sequence $\langle 1, 2, 3, 4, 5, \dots \rangle$?

The generating function is $1 + 2x + 3x^2 + 4x^3 + 5x^4 + \dots$

How to obtain a closed form of this function?

$$\begin{aligned} \frac{d}{dx}(1 + x + x^2 + x^3 + \dots) &\Leftrightarrow \frac{d}{dx}\left(\frac{1}{1-x}\right) \\ 1 + 2x + 3x^2 + 4x^3 + \dots &\Leftrightarrow \frac{1}{(1-x)^2} \\ \langle 1, 2, 3, 4, 5, \dots \rangle &\Leftrightarrow \frac{1}{(1-x)^2} \end{aligned}$$

Differentiation (2/2)

How to generate the sequence $\langle 1, 4, 9, 16, \dots \rangle$?

$$1 + 2x + 3x^2 + 4x^3 + \dots \Leftrightarrow \frac{1}{(1-x)^2}$$

$$x + 2x^2 + 3x^3 + 4x^4 + \dots \Leftrightarrow \frac{x}{(1-x)^2}$$

$$\frac{d}{dx}(x + 2x^2 + 3x^3 + 4x^4 + \dots) \Leftrightarrow d\left(\frac{x}{(1-x)^2}\right)$$

$$= 1 + 4x + 9x^2 + 16x^3 + \dots \Leftrightarrow \frac{1+x}{(1-x)^3}$$

$$\langle 1, 4, 9, 16, \dots \rangle \Leftrightarrow \frac{1+x}{(1-x)^3}$$

Product

$$\langle a_0, a_1, a_2, a_3, \dots \rangle \Leftrightarrow A(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$$

$$\langle b_0, b_1, b_2, b_3, \dots \rangle \Leftrightarrow B(x) = b_0 + b_1x + b_2x^2 + b_3x^3 + \dots$$

What is the sequence corresponds to the polynomial $C(x) = A(x)B(x)$?

$$C(x) = A(x)B(x) = (a_0 + a_1x + \dots)(b_0 + b_1x + \dots) = \sum_{k=0}^{\infty} c_k x^k$$

$$c_k = a_0b_k + a_1b_{k-1} + a_2b_{k-2} + \dots + a_{k-1}b_1 + a_kb_0 = \sum_{i=0}^k a_ib_{k-i}$$

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General strategy

Coefficient of x^k = number of ways to choose k items.

$$\left\langle \binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{k}, 0, 0, \dots \right\rangle \Leftrightarrow \binom{n}{0} + \binom{n}{1}x + \dots + \binom{n}{k}x^n = (1+x)^n$$

the coefficient of x^k in $(1+x)^n$ is the number of ways to choose k distinct items from a set of size n .

Convolution rule

1. Let $A(x)$ be the generating function for selecting items from set A .
2. Let $B(x)$ be the generating function for selecting items from set B .
3. If A and B are disjoint, then the generating function for selecting items from the union $A \cup B$ is the product $A(x) \cdot B(x)$.

$$A(x) = a_0 + a_1x + a_2x^2 + \dots$$

$$B(x) = b_0 + b_1x + b_2x^2 + \dots$$

$$c_k = a_0b_k + a_1b_{k-1} + a_2b_{k-2} + \dots + a_{k-1}b_1 + a_kb_0 = \sum_{i=0}^k a_ib_{k-i}$$

$$C(x) = A(x)B(x) = (a_0 + a_1x + \dots)(b_0 + b_1x + \dots)$$

Example: choosing fruits: an “impossible” counting problem...

How many ways can we fill a bag with n fruits with the following constraints?

- The number of apples must be even.
- The number of bananas must be a multiple of 5.
- There can be at most one mango.
- There can be at most four oranges.

For example, when $n = 6$, there are 7 ways to form a bag with 6 fruits.

Apple	6	4	4	2	2	0	0
Banana	0	0	0	0	0	5	5
Mango	0	0	1	0	1	0	1
Orange	0	2	1	4	3	1	0

Example: choosing fruits: generating functions (1/2)

- The number of apples must be even.
- The number of bananas must be a multiple of 5.
- There can be at most one mango.
- There can be at most four oranges.

GF for **a**pples:

$$A(x) = x^0 + x^2 + x^4 + \dots = \frac{1}{1 - x^2}$$

GF for **b**ananas:

$$B(x) = x^0 + x^5 + x^{10} + \dots = \frac{1}{1 - x^5}$$

GF for **m**angoes:

$$M(x) = x^0 + x^1 = 1 + x$$

GF for **o**ranges:

$$O(x) = x^0 + x^1 + x^2 + x^3 + x^4 = \frac{1 - x^5}{1 - x}$$

GF for fruits:

$$A(x)B(x)M(x)O(x) = \frac{1}{1 - x^2} \cdot \frac{1}{1 - x^5} \cdot (1 + x) \cdot \frac{1 - x^5}{1 - x} = \frac{1}{(1 - x)^2}$$

Example: choosing fruits: generating functions (2/2)

Generating function for fruits:

$$\frac{1}{(1-x)^2}$$

How many ways can we fill a bag with n fruits with the following constraints?

$$\frac{d}{dx}(1 + x + x^2 + x^3 + \dots) \Leftrightarrow \frac{d}{dx}\left(\frac{1}{1-x}\right)$$

$$1 + 2x + 3x^2 + 4x^3 + \dots \Leftrightarrow \frac{1}{(1-x)^2}$$

The number of ways to choose n fruits is the coefficient of x^n , which is exactly $n + 1$

Exercises (1/2)

1. How many ways are there to get n fruits that satisfy the following three conditions?
 - There are at most 2 apples.
 - The number of bananas is arbitrary.
 - The number of mangoes must be divisible by 3.
2. How many ways can we select n mangoes with k varieties?
3. How many ways can we select n mangoes with k varieties such that each variety has at least two?

Exercises (2/2)

4. We need \$17 to contribute to the disaster relief for the people affected by the flood.

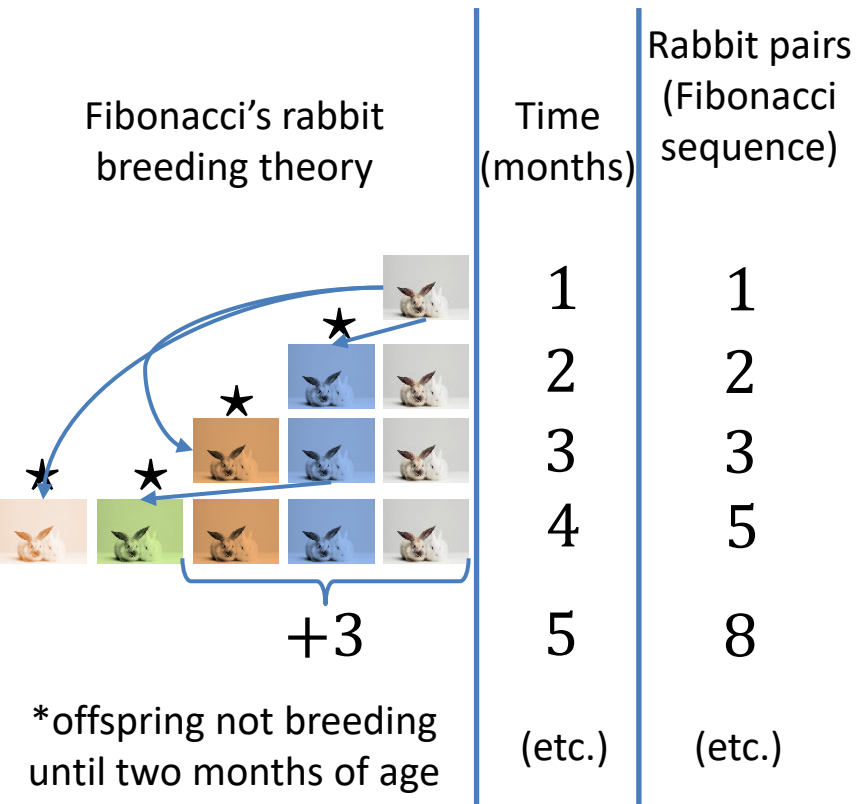
- There are 20 students participating in the contribution.
- It is known that the first 19 people will contribute \$1 or nothing, and the 20th person will contribute \$1, \$5, or nothing.

Use a generating function to calculate the number of ways to contribute \$17.

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Example: Fibonacci in Animal Breeding Patterns



1. Rabbits follow the Fibonacci sequence in their reproductive pattern.
2. A single pair of rabbits (one male and one female) produces another pair in the second month.
3. From the third month onwards, each pair produces a new pair every month.
4. The sequence of pairs of rabbits is 1, 1, 2, 3, 5, 8, 13, and so on.
5. The breeding pattern reflects the natural balance of reproduction and population growth.
6. The population grows slowly initially as the breeding pairs are young and need time to mature.
7. As the population increases, more pairs are available to reproduce, leading to an exponential growth rate that follows the Fibonacci sequence.

Rabbit Populations

- Rabbits mature after one month, i.e., an offspring does not breed until two months of age.
 - w_n = # newborn pairs after n months
 - r_n = # reproducing pairs after n months
- Start with a newborn pair: $w_0 = 1, r_0 = 0$
- $r_1 = 1$
- $r_n = r_{n-1} + w_{n-1}$
- $w_n = r_{n-1}$ SO
- $r_n = r_{n-1} + r_{n-2}$

How many rabbits after n months?

It was Fibonacci who was studying rabbit population growth.

Fibonacci sequence

- The Fibonacci sequence we want to analyze is:

$$\langle r_0, r_1, r_2, r_3, \dots \rangle = \langle 0, 1, 1, 2, 3, 5, 8, 11, \dots \rangle$$

- Define a generating function for this sequence:

$$R(x) = r_0 + r_1x + r_2x^2 + r_3x^3 + \dots$$

- Remember $r_{i+2} = r_{i+1} + r_i$

First we want to obtain a closed form for $R(x)$

Generating Function for Rabbits

$$\begin{array}{rcl}
 R(x) & = & r_0 + r_1x + r_2x^2 + r_3x^3 + \dots \\
 -xR(x) & = & -r_0x - r_1x^2 - r_2x^3 - r_3x^4 - \dots \\
 -x^2R(x) & = & -r_0x^2 - r_1x^3 - r_2x^4 - r_3x^5 - \dots
 \end{array}$$

0
0
0
...

Remember $r_{i+2} = r_{i+1} + r_i$

Closed form of $R(x)$ (1/2)

$$\begin{aligned}
 R(x) &= r_0 + r_1x + r_2x^2 + r_3x^3 + \dots \\
 -xR(x) &= -r_0x - r_1x^2 - r_2x^3 - r_3x^4 - \dots \\
 -x^2R(x) &= -r_0x^2 - r_1x^3 - r_2x^4 - r_3x^5 - \dots
 \end{aligned}$$

$$R(x) - xR(x) - x^2R(x) = r_0 + r_1x - r_0x = x$$



$$R(x) = \frac{x}{1 - x - x^2}$$

Closed form of $R(x)$ (2/2)

$$R(x) = \frac{x}{1 - x - x^2}$$

What is the closed form of r_n ?

- $R(x) = \frac{x}{(1-\alpha x)(1-\beta x)} = \frac{A}{1-\alpha x} + \frac{B}{1-\beta x}$
- $\frac{1}{1-\alpha x} = 1 + \alpha x + \alpha^2 x^2 + \dots + \alpha^n x^n + \dots$
- $\frac{1}{1-\beta x} = 1 + \beta x + \beta^2 x^2 + \dots + \beta^n x^n + \dots$
- So $r_n = \text{coefficient of } x^n \text{ in } R(x) = A\alpha^n + B\beta^n$

Closed form of coefficients

- $$R(x) = \frac{x}{(1-\alpha x)(1-\beta x)} = \frac{A}{1-\alpha x} + \frac{B}{1-\beta x}$$

$$\alpha = \frac{1+\sqrt{5}}{2} \text{ and } \beta = \frac{1-\sqrt{5}}{2}$$

$$A = \frac{1}{\alpha-\beta} = \frac{1}{\sqrt{5}} \text{ and } B = \frac{1}{\beta-\alpha} = -\frac{1}{\sqrt{5}}$$
- So r_n = coefficient of x^n in $R(x)$

$$= A\alpha^n + B\beta^n$$

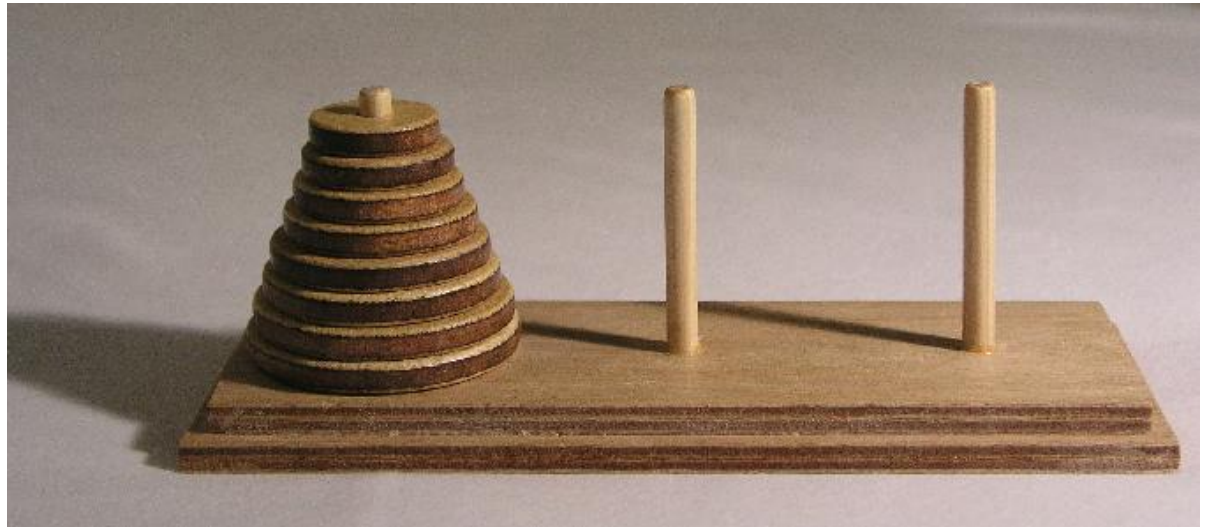
$$= \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

Hanoi's Tower

How to solve these recurrences without using techniques like the master theorem?

The objective of the puzzle is to move the entire stack to one of the other rods, obeying the following rules:

1. Only one disk may be moved at a time.
2. Each move consists of taking the upper disk from one of the stacks and placing it on top of another stack or an empty rod.
3. No disk may be placed on top of a disk that is smaller than it.



Number of moves:

$$T(n) = 2T(n - 1) + 1$$

Generating Function

- m_n : number of moves with n disks.

- $m_0 = 0, m_n = 2m_{n-1} + 1$.

- The sequence we want to analyze is:

$$m_0, m_1, m_2, m_3, \dots$$

- Define a generating function for this sequence:

$$M(x) = m_0 + m_1x + m_2x^2 + m_3x^3 + \dots$$

First we want to obtain a closed form for $M(x)$

Generating Function for Hanoi's Tower

$$\begin{array}{rcl}
 M(x) & = & m_0 + m_1x + m_2x^2 + m_3x^3 + \dots \\
 -2xM(x) & = & -2m_0x - 2m_1x^2 - 2m_2x^3 - \dots \\
 -G(x) & = & -1 - x - x^2 - x^3 - \dots
 \end{array}$$

0 0 0 ...

Remember $m_{i+1} = 2m_i + 1$

Generating Function for Hanoi's Tower

$$\begin{array}{rcl}
 M(x) & = & m_0 + m_1x + m_2x^2 + m_3x^3 + \dots \\
 -2xM(x) & = & -2m_0x - 2m_1x^2 - 2m_2x^3 - \dots \\
 -G(x) & = & -1 - x - x^2 - x^3 - \dots
 \end{array}$$

$$M(x) - 2xM(x) - G(x) = m_0 - 1 = -1$$

Remember $G(x) = \frac{1}{1-x}$



$$R(x) = \frac{-1 + G(x)}{1 - 2x} = \frac{-1 + \frac{1}{1-x}}{1 - 2x} = \frac{x}{(1-x)(1-2x)}$$

Closed form of $M(x)$ (2/2)

$$M(x) = \frac{x}{(1-x)(1-2x)}$$

What is the closed form of m_n ?

- $M(x) = \frac{x}{(1-x)(1-2x)} = \frac{-1}{1-x} + \frac{1}{1-2x}$

- $\frac{1}{1-x} = 1 + x + x^2 + \dots + x^n + \dots$

- $\frac{1}{1-2x} = 1 + 2x + 2^2x^2 + \dots + 2^nx^n + \dots$

- So $m_n = \text{coefficient of } x^n \text{ in } M(x) = -1 + 2^n$

Exercise

Consider the sequence a_n defined by the recurrence relation:

$$\begin{cases} a_0 = 1 \\ a_n = 2a_{n-1} + 3 \end{cases}$$

1. Derive the ordinary generating function $A(x)$ for the sequence a_n .
2. Use the generating function to find a closed-form expression for a_n .

Q & A